THE DIAGONAL OF THE COHERENT SHEAVES STACK IS REPRESENTABLE BY ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Situation situation-coherent (canonical tag 08KB). cited at statement line 1170. Standing situation.
- Lemma algebraic-lemma-representable-diagonal (canonical tag 045G). cited at proof line 1180. The diagonal criterion via Isom functors.
- Proposition proposition-isom (canonical tag 08K9). cited at proof line 1192. Isom(F, G) is an algebraic space.

Source reference. Original definition 08KB: sources/stacks-chow/tags.txt, source lines 10881–10881. Cited at statement line 1170: standing situation.

Source reference. Original lemma 045G: sources/stacks-chow/tags.txt, source lines 5435–5435. Cited at proof line 1180: the diagonal criterion.

Source reference. Original proposition 08K9: sources/stacks-chow/tags.txt, source lines 10877–10877. Cited at proof line 1192: Isom(F, G) is an algebraic space.

Lemma 0.2. *In Situation 08KB. Denote $\mathcal{X} = \mathcal{C}oh_{X/B}$. Then $\Delta : \mathcal{X} \to \mathcal{X} \times \mathcal{X}$ is representable by algebraic spaces.*

Proof. Consider two objects $x = (T, g, \mathcal{F})$ and $y = (T, h, \mathcal{G})$ of $\mathcal{X}$ over a scheme T. We have to show that $Isom_{\mathcal{X}}(x, y)$ is an algebraic space over T, see Algebraic Stacks, Lemma 045G. If for $a : T' \to T$ the restrictions $x|_{T'}$ and $y|_{T'}$ are isomorphic in the fibre category $\mathcal{X}_{T'}$, then $g \circ a = h \circ a$. Hence there is a transformation of presheaves

$$Isom_{\mathcal{X}}(x, y) \longrightarrow \text{Equalizer}(g, h)$$

Since the diagonal of B is representable (by schemes) this equalizer is a scheme. Thus we may replace T by this equalizer and the sheaves $\mathcal{F}$ and $\mathcal{G}$ by their pullbacks. Thus we may assume $g = h$. In this case we have $Isom_{\mathcal{X}}(x, y) = Isom(\mathcal{F}, \mathcal{G})$ and the result follows from Proposition 08K9. □

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